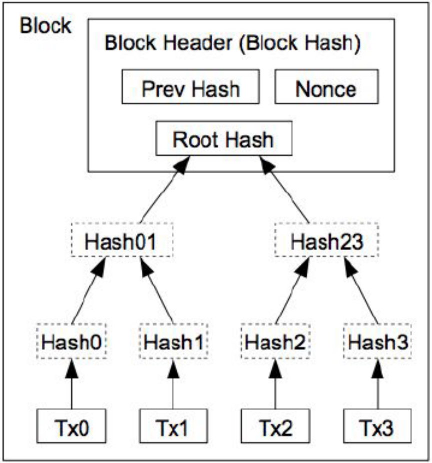
IT486: simplified payment verification (spv)

Structure of a block

- a header (80 bytes) consisting of three sets of metadata
  - a hash of the block header of the previous (last added) block of the blockchain
  - information related to PoW
  - root of the Merkle tree summarizing all the transactions in the block
- a list of transactions that make up the bulk of its size
  - almost 1000 times larger than the block header

Structure of a block



Transactions Hashed in a Merkle Tree

Fully-validating nodes

- Fully validate transactions and blocks, and relay them to other nodes
- Store entire block chain

Thin clients

- Thin clients are more viable for less powerful/mobile devices
  - also called Simplified Payment Verification (SPV) nodes
  - download a copy of the headers of all blocks by querying a full node
  - check only transactions they care about by querying a full node

SPV: Verifying transactions

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- The full node responds with
  - the block header hash of the block the transaction is included in
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- The Merkle root is also part of the block header, so the SPV node already has that too

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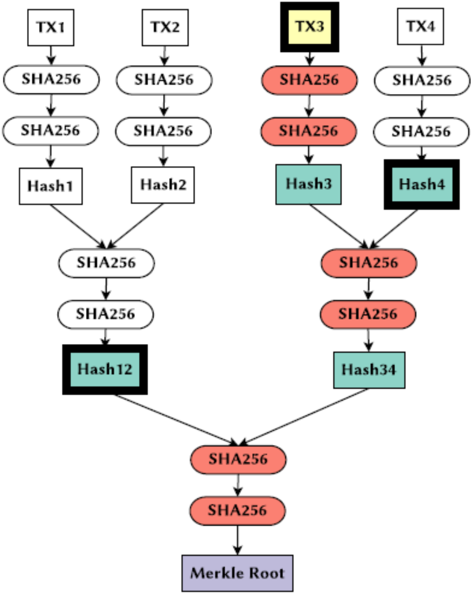

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- The SPV node needs to rehash the txn and verify the Merkle root that is stored in the block header
- Using the Merkle path, it only downloads the relevant hashes from the full node
- The SPV node does not have to know anything about the other transactions in the block

SPV: Verifying transactions

- A full node can't lie and say that a transaction existed in a block when it didn't actually exist
- But it can lie by saying that a transaction that does exist in a block did not happen

Privacy issue

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- Bloom filters have been to added to SPV clients to mitigate this risk

Recap: Bloom Filters

- Bloom filters are data structures that allow you to:
  - insert an element into a set
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  - query whether an element is in a set
- When the answer to the query is "yes", it can be wrong (false positive!)
- "no" answers are always correct (zero false negatives)

How SPV nodes use bloom filters

- The SPV node
  - creates an empty bloom filter
  - makes a list of everything it is interested in
    - addresses, keys, hashes
  - adds all of these to the bloom filter
- The SPV node sends the bloom filter to the full node

How SPV nodes use bloom filters

- The SPV node
  - creates an empty bloom filter
  - makes a list of everything it is interested in
    - addresses, keys, hashes
  - adds all of these to the bloom filter
- The SPV node sends the bloom filter to the full node
- The full node checks the bloom filter against the blockchain
  - only txns that match the bloom filter are sent back to the SPV node

How SPV nodes use bloom filters

- For each matching txn, the full node sends back
  - block headers, Merkle paths, txn messages
- The SPV node
  - verifies the info of relevant transactions
  - discards all other info